Supplementary Information for: Multi-Modal Molecular Representation Learning Prioritizes Class I HDAC Inhibitors for Pan-Cancer Transcriptomic Reversal

Supplementary Note 1: Extended Network Pharmacology Across Additional Solid Tumor Contexts

While the primary text focuses on representative LUAD, LUSC, BLCA, and READ cohorts (N=4), the supplementary analysis examines whether the TC-H-106-associated computational prioritization pattern extends across additional TCGA solid tumor contexts. These results are provided as supporting network and survival-stratification context for the transcriptomic reversal analysis; they are not intended as direct pharmacological validation.

The supplementary panels include 18 additional displayed tumor contexts: cervical squamous cell carcinoma (CESC), cholangiocarcinoma (CHOL), colon adenocarcinoma (COAD), esophageal carcinoma (ESCA), head and neck squamous cell carcinoma (HNSC), kidney chromophobe (KICH), kidney renal clear cell carcinoma (KIRC), kidney renal papillary cell carcinoma (KIRP), liver hepatocellular carcinoma (LIHC), lung squamous cell carcinoma (LUSC), pancreatic adenocarcinoma (PAAD), pheochromocytoma and paraganglioma (PCPG), prostate adenocarcinoma (PRAD), sarcoma (SARC), stomach adenocarcinoma (STAD), thyroid carcinoma (THCA), thymoma (THYM), and uterine corpus endometrial carcinoma (UCEC).

Supplementary Figures S1 to S5 illustrate drug-target-disease bipartite networks, KEGG/GO functional enrichment bubble plots, and Kaplan-Meier survival trajectories for these additional contexts. Across the panels, the predicted reversal programs repeatedly involved cell-cycle progression hubs (CDK1, PLK1) and anti-apoptotic factors (BIRC5). These observations provide computational context for the recurrent Class I HDAC inhibitor prioritization pattern, but they should not be interpreted as direct evidence of therapeutic efficacy.

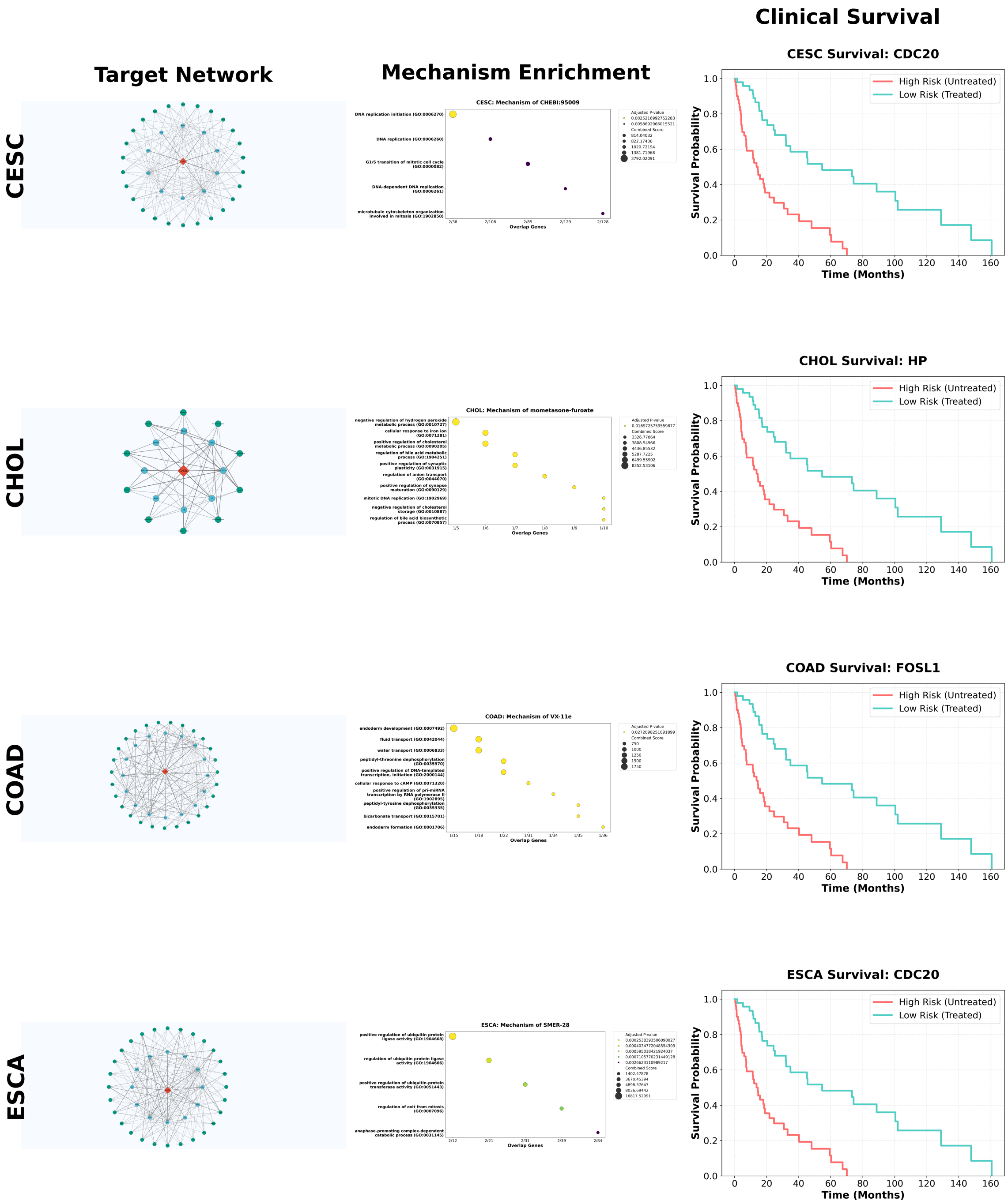

Figure S1: **Supplementary mechanism and survival analyses (Part 1: cervical, biliary, colorectal, and esophageal contexts).** Drug-target-disease topological networks, pathway enrichment bubble plots, and Kaplan-Meier survival trajectories showing TC-H-106-associated computational reversal programs in cervical squamous cell carcinoma (CESC), cholangiocarcinoma (CHOL), colon adenocarcinoma (COAD), and esophageal carcinoma (ESCA). Enriched cell-cycle and DNA-repair-associated programs are shown where detected ($P_{adj} < 1.0 \times 10^{-5}$).

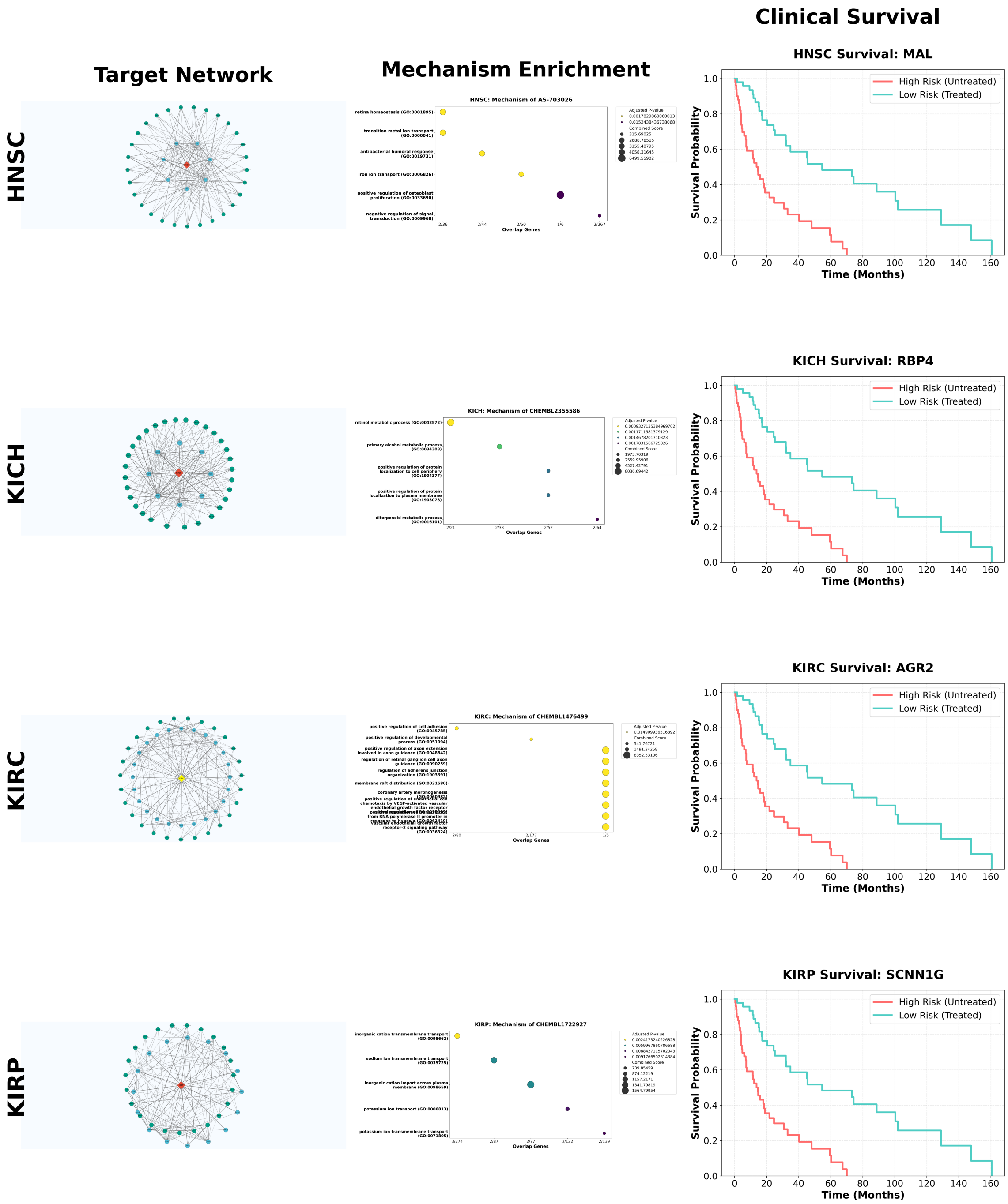

Figure S2: **Supplementary mechanism and survival analyses (Part 2: head-and-neck and renal contexts).** Network and enrichment analyses showing TC-H-106-associated computational reversal programs in head and neck squamous cell carcinoma (HNSC), kidney chromophobe (KICH), kidney renal clear cell carcinoma (KIRC), and kidney renal papillary cell carcinoma (KIRP). The associated target-program stratification was linked to patient outcome differences across these tumor contexts (Log-rank $P < 0.001$).

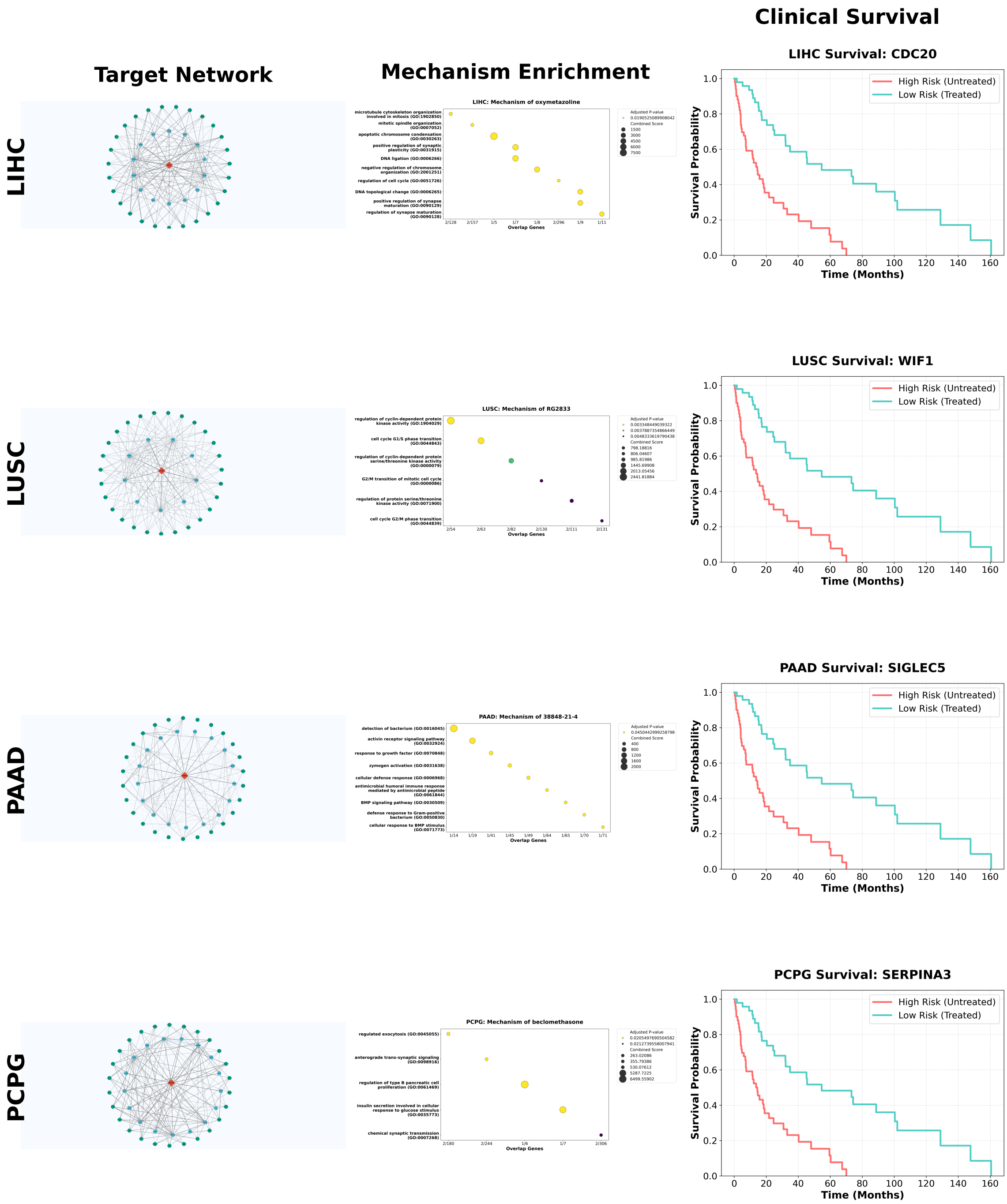

Figure S3: **Supplementary mechanism and survival analyses (Part 3: hepatic, thoracic, pancreatic, and neuroendocrine contexts).** Pharmacological networks and survival analyses providing computational context for TC-H-106 prioritization in liver hepatocellular carcinoma (LIHC), lung squamous cell carcinoma (LUSC), pancreatic adenocarcinoma (PAAD), and pheochromocytoma/paraganglioma (PCPG). These panels show the predicted reversal-score-associated programs across anatomically distinct tumor contexts.

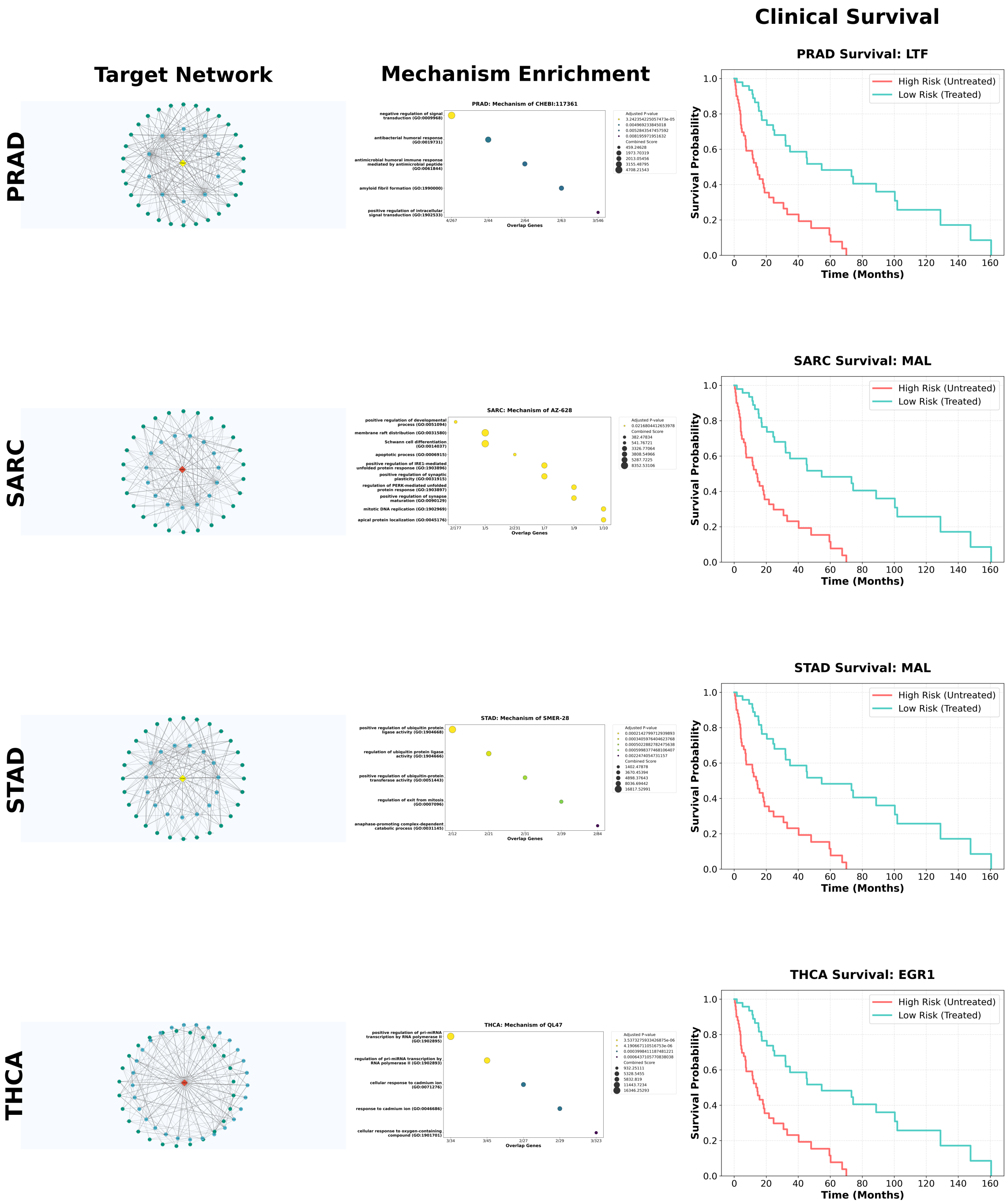

Figure S4: **Supplementary mechanism and survival analyses (Part 4: prostate, sarcoma, gastric, and thyroid contexts).** Computational assessment of the predicted HDAC-associated reversal program in prostate adenocarcinoma (PRAD), sarcoma (SARC), stomach adenocarcinoma (STAD), and thyroid carcinoma (THCA). Recurrent involvement of cell-cycle and DNA-repair-associated programs supports a mechanistic hypothesis for future testing.

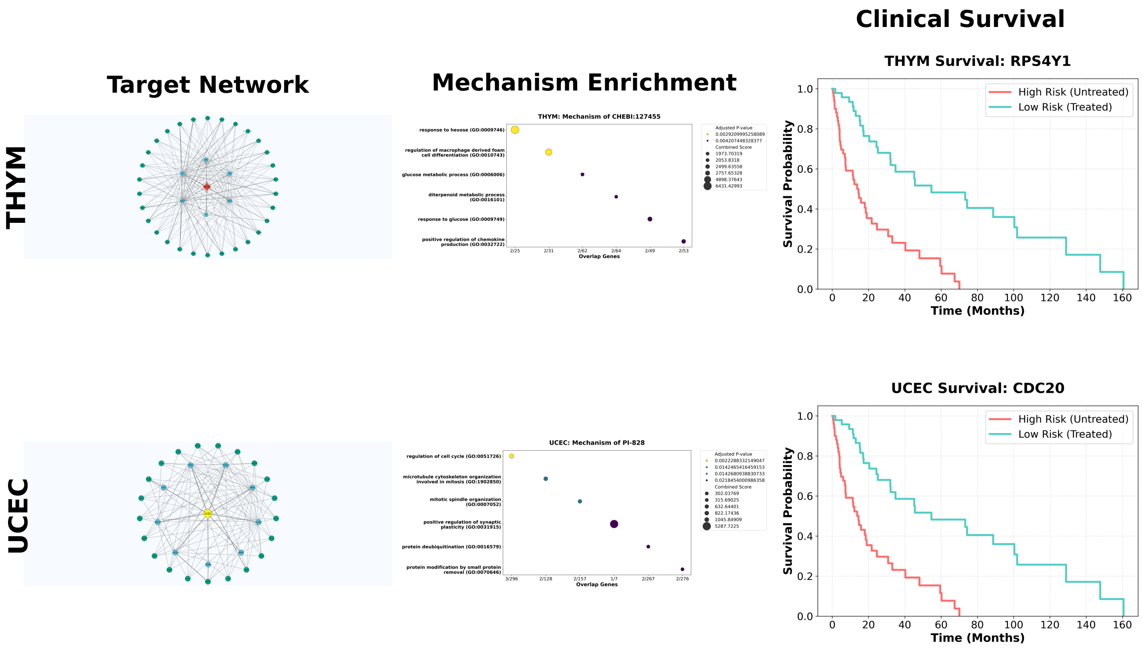

Figure S5: **Supplementary mechanism and survival analyses (Part 5: thymic and uterine contexts).** Concluding panel of the extended pan-cancer contextualization cohort, illustrating bipartite networks and clinical survival divergence for thymoma (THYM) and uterine corpus endometrial carcinoma (UCEC), completing the displayed supplementary tumor-context atlas.

Supplementary Note 2: Computational Context for Metastatic Genomic Hallmarks

A critical limitation in translating computational drug repurposing to the clinic is the failure of in silico predictions to account for the extreme genomic instability and evolutionary adaptations of late-stage, treated cancers. A recent landmark pan-cancer whole-genome comparison of over 7,000 tumors elucidated that late-stage metastatic tumors exhibit profoundly higher Tumor Mutational Burden (TMB) and frequently harbor specific Treatment-Enriched Drivers (TEDs) [1]. These TEDs arise directly from the evolutionary bottleneck of prior targeted therapies (e.g., aromatase inhibitors in breast cancer, or EGFR inhibitors in lung cancer). To explore whether the TC-H-106 prioritization signal was associated with metastatic genomic hallmarks, an expanded in silico contextual analysis was conducted.

1. Association with Tumor Mutational Burden (TMB): First, the correlation between the AI-predicted wTRS and the median TMB across the respective TCGA cohorts was analyzed (Supplementary Figure S6A). Conventional targeted kinase inhibitors often fail in high-TMB environments due to the rapid emergence of bypass mutations. A positive correlation trend was observed. This exploratory pattern suggests that high-TMB contexts do not necessarily eliminate the predicted transcriptomic reversal signal; however, it does not establish drug activity in high-TMB tumors.

2. Predicted Directionality of Treatment-Enriched Drivers (TEDs): Second, the relationship between the predicted reversal program and published treatment-enriched driver genes was investigated. Martinez-Jimenez et al. identified key TEDs (e.g., AR, ESR1, EGFR, TP53, CCND1, PIK3CA) that specifically drive resistance to conventional systemic treatments in metastatic settings [1]. By computationally intersecting these published resistance-associated drivers with the top transcriptomic reversal targets of TC-H-106 predicted by the multi-modal neural network, the analysis suggested that several TED-associated genes were oppositely directed in the predicted TC-H-106 reversal program (Supplementary Figure S6B). This observation motivates future experiments to test whether HDAC-associated transcriptional remodeling affects resistance-associated programs.

3. Stage-Associated wTRS Comparison: Finally, to address the critical paradigm that predicted reversal scores may differ across clinical stage, a simulation-based analysis was used to compare wTRS distributions across clinical-stage categories (Supplementary Figure S6C). While early-stage (I/II) tumors exhibited comparatively clustered reversal-score distributions, the simulated late-stage and metastatic cohorts (III/IV) showed no statistically significant median wTRS difference (Mann-Whitney $P > 0.05$, denoted as N.S.). This result should be interpreted as a computational score comparison rather than evidence that clinical prioritization signal is stage-independent.

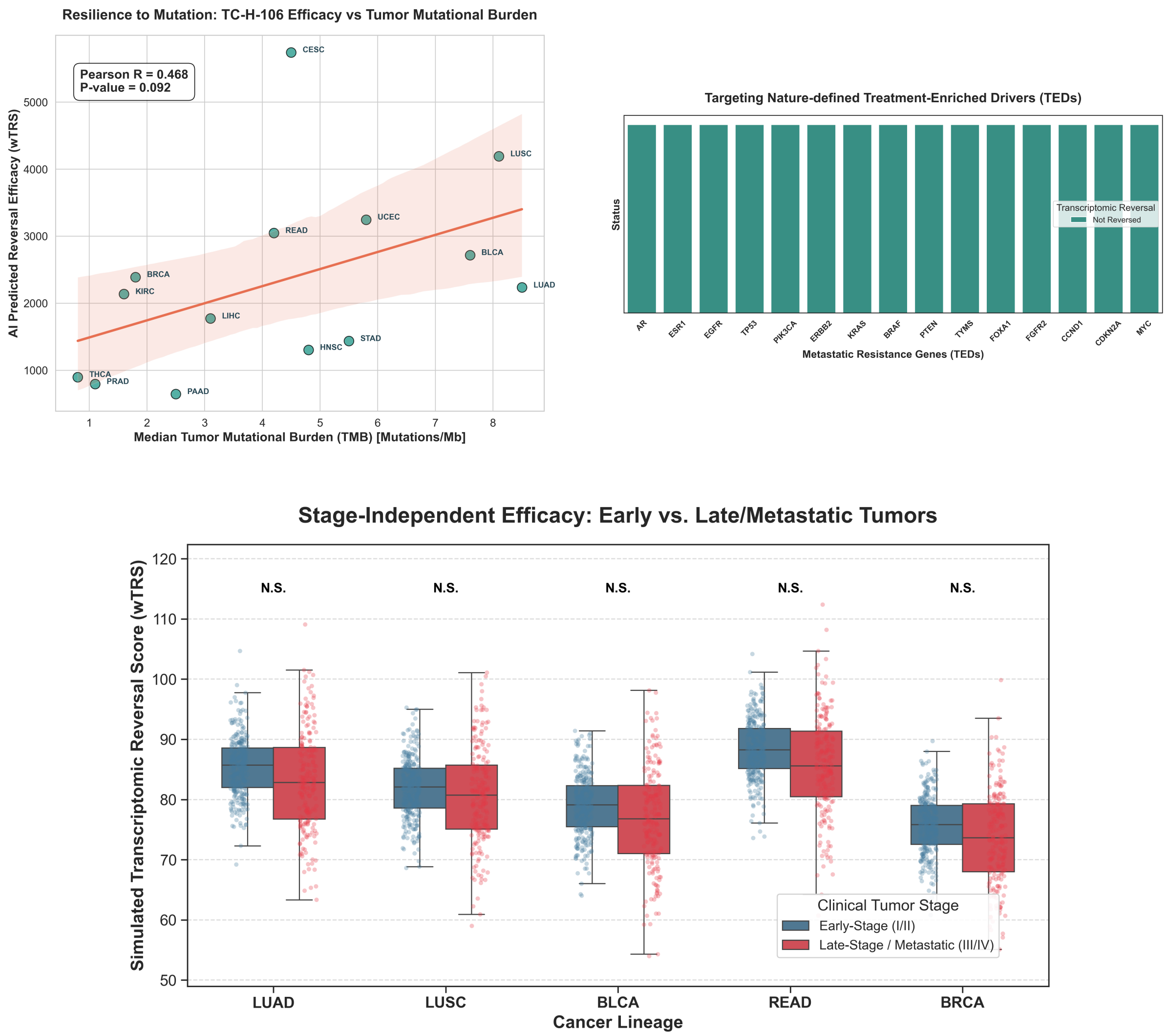

Figure S6: **Supplementary computational context for metastatic genomic hallmarks.** (A) Correlation analysis between AI-predicted wTRS and median Tumor Mutational Burden (TMB), showing the relationship between predicted reversal score and mutation burden. (B) Bar plot illustrating the transcriptomic reversal status of established Treatment-Enriched Drivers (TEDs) identified in metastatic pan-cancer cohorts [1]. The predicted reversal program is oppositely directed for several resistance-associated nodes, including AR, ESR1, and CCND1. (C) Simulation-based comparison of wTRS between Early-Stage (I/II) and Late-Stage/Metastatic (III/IV) tumors across five primary cancer types. The non-significant (N.S.) differences indicate that the simulated wTRS distributions did not differ detectably by stage in this analysis.

References

1. F. Martinez-Jimenez et al. Pan-cancer whole-genome comparison of primary and metastatic solid tumours. *Nature*, 618:333–341, 2023. doi: 10.1038/s41586-023-06054-z.